

The multimode conditional quantum Entropy Power Inequality and the squashed entanglement of the multimode extreme bosonic Gaussian channels

Alessandro Falco[✉] and Giacomo De Palma[✉]

Abstract—We prove the multimode conditional quantum Entropy Power Inequality for bosonic quantum systems. This inequality determines the minimum conditional von Neumann entropy of the output of the most general linear mixing of bosonic quantum modes among all the input states of the modes with given conditional entropies. Bosonic quantum systems constitute the mathematical model for the electromagnetic radiation in the quantum regime, which provides the most promising platform for quantum communication and quantum key distribution. We apply our multimode conditional quantum Entropy Power Inequality to determine new lower bounds to the squashed entanglement of a large family of bosonic Gaussian states. The squashed entanglement is one of the main entanglement measures in quantum communication theory, providing the best known upper bound to the distillable key. Exploiting this result, we determine a new lower bound to the squashed entanglement of the multimode bosonic Gaussian channels that are extreme, *i.e.*, that cannot be decomposed as a non-trivial convex combination of quantum channels. The squashed entanglement of a quantum channel provides an upper bound to its secret-key capacity, *i.e.*, the capacity to generate a secret key shared between the sender and the receiver. Lower bounds to the squashed entanglement are notoriously hard to prove. Our results contribute to break this barrier and will stimulate further research in the field of quantum communication with bosonic quantum systems.

Index Terms—Bosonic Gaussian channels, continuous variables, Entropy Power Inequality, squashed entanglement.

I. INTRODUCTION

THE Shannon differential entropy of a random variable X with values in $\mathbb{R}^n$ and probability density $p(\mathbf{x})d^n\mathbf{x}$ is [1]

$$S(X) := - \int_{\mathbb{R}^n} \ln(p(\mathbf{x})) dp(\mathbf{x}), \quad (1)$$

and quantifies the noise or the information contained in X . The classical Entropy Power Inequality (EPI) [2]–[4] determines the minimum Shannon differential entropy of the sum of two independent random variables X_1 and X_2 with values in $\mathbb{R}^n$ and given Shannon differential entropies:

$$\exp\left(\frac{2S(X_1 + X_2)}{n}\right) \geq \exp\left(\frac{2S(X_1)}{n}\right) + \exp\left(\frac{2S(X_2)}{n}\right), \quad (2)$$

Alessandro Falco was with the Scuola Normale Superiore, 56127 Pisa, Italy. He is now with the Institute for Quantum Inspired and Quantum Optimization, Technische Universität Hamburg, Germany and also with the Quantum Research Centre, Technology Innovation Institute, Abu Dhabi, United Arab Emirates (e-mail: alessandro.falco@sns.it).

Giacomo De Palma is with the Department of Mathematics, University of Bologna, Bologna, Italy (e-mail: giacomo.depalma@unibo.it).

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and is a fundamental element of classical information theory [1]. For any random variable X with values in $\mathbb{R}^n$ and any invertible $M \in \mathbb{R}^{n \times n}$ we have

$$S(MX) = S(X) + \frac{1}{2} \ln(\det(M)). \quad (3)$$

Let $X_1, \dots, X_K$ be independent random variables taking values in $\mathbb{R}^n$, and let $M_1, \dots, M_K \in \mathbb{R}^{n \times n}$. Let us consider the linear combination

$$Y := \sum_{k=1}^K M_k X_k. \quad (4)$$

The property (3) implies the following multimode generalization of the EPI (2):

$$\exp\left(\frac{2S(Y)}{n}\right) \geq \sum_{k=1}^K |\det(M_k)|^{\frac{2}{n}} \exp\left(\frac{2S(X_k)}{n}\right). \quad (5)$$

Let now Z be an arbitrary random variable, and let $X_1, \dots, X_K$ be random variables conditionally independent given Z and with values in $\mathbb{R}^n$. The entropy of the random variable X conditioned on the random variable Z is the expectation value with respect to Z of the entropy of X given the value of Z :

$$S(X|Z) = \int S(X|Z=z) dp(z). \quad (6)$$

Jensen's inequality implies the following conditional version of the EPI (5):

$$\exp\left(\frac{2S(Y|Z)}{n}\right) \geq \sum_{k=1}^K |\det(M_k)|^{\frac{2}{n}} \exp\left(\frac{2S(X_k|Z)}{n}\right). \quad (7)$$

The first result of this paper is to prove a quantum generalization of the EPI (7). The quantum counterpart of probability measures are quantum states, which are positive-semidefinite linear operators with unit trace acting on the Hilbert space of a quantum system. The counterpart of the probability measures on $\mathbb{R}^n$ with even n are the states of a bosonic quantum system with $m = n/2$ modes, and the counterpart of the coordinates of $\mathbb{R}^n$ are self-adjoint linear operators called quadratures. Bosonic quantum systems [5]–[9], model electromagnetic waves in the quantum regime. Electromagnetic waves traveling through optical fibers or free space provide the most promising platform for quantum communication and quantum key distribution [10]. Bosonic quantum systems then play a key role in quantum communication and quantum cryptography and provide the model to determine the maximum communication and key distribution rates achievable in